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# =====
# MANEIRA 1 — usando math.log()
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import math
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```
r1 = math.log(4**3, 2)
r2 = math.log(math.sqrt(27), 3)
r3 = math.log(25 * math.sqrt(5), 5)
r4 = math.log(64, 0.5)
r5 = math.log10(0.001)
r6 = math.log(1/6, 6)
r7 = math.log(2, 8)
r8 = 6**(math.log10(1))
r9 = math.log(math.e**52)
```

```
L = (math.log10(1) + math.log(math.e) - math.log10(10)) / \
    (1 - math.log10(0.01) + math.log10(math.sqrt(10)))
```

```
print(r1, r2, r3, r4, r5, r6, r7, r8, r9, L)
```

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# =====
# MANEIRA 2 — propriedades algébricas
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r1 = 3          # log2(4³) = log2(2⁶) = 6
r1 = 6
```

```
r2 = 3/2        # √27 = 3^(3/2)
```

```
r3 = 2 + 1/2    # 25√5 = 5² * 5^(1/2)
```

```
r4 = -6        # (1/2)^(-6) = 64
```

```
r5 = -3        # 10^-3 = 0,001
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```
r6 = -1        # 6^-1 = 1/6
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```
r7 = 1/3       # 8^(1/3)=2
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```
r8 = 1         # log(1)=0 → 6^0=1
```

```
r9 = 52        # ln(e^52)=52
```

```

L = 0                      # numerador = 0

print(r1, r2, r3, r4, r5, r6, r7, r8, r9, L)

# =====
# MANEIRA 3 — usando sympy
# =====

from sympy import *

r1 = log(4**3, 2)
r2 = log(sqrt(27), 3)
r3 = log(25*sqrt(5), 5)
r4 = log(64, Rational(1,2))
r5 = log(Rational(1,1000), 10)
r6 = log(Rational(1,6), 6)
r7 = log(2, 8)
r8 = 6**(log(1,10))
r9 = ln(E**52)

L = (log(1,10) + ln(E) - log(10,10)) / \
    (1 - log(Rational(1,100),10) + log(sqrt(10),10))

print(simplify(r1))
print(simplify(r2))
print(simplify(r3))
print(simplify(r4))
print(simplify(r5))
print(simplify(r6))
print(simplify(r7))
print(simplify(r8))
print(simplify(r9))
print(simplify(L))

```